

Research Methods and Stats for Psychologists

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Overview

- Welcome!
- What this module is, how it runs, and how you'll be assessed

Fake news

- What is 'true' in psychological science?
- Parents and educators may believe 'myths' about child development without knowing whether they are supported by evidence.
- As scientists, it's important to promote evidence-based practices and debunk false assumptions.



Myth or fact?

“Showing cognitively stimulating videos to babies boosts their development?”

Myth!

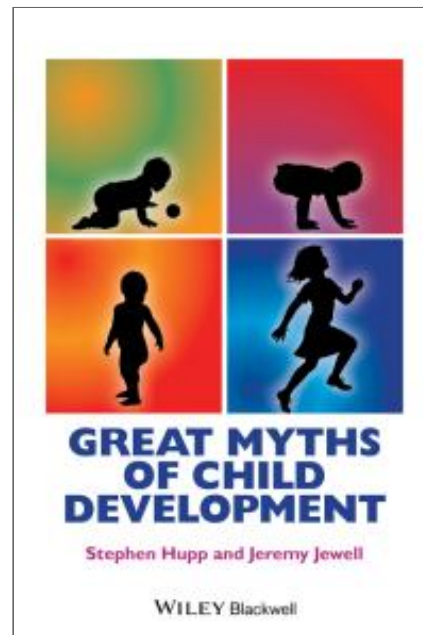
- Watching a DVD 15 times over 6 weeks did not significantly increase word learning in 12- and 15-month-olds (Robb et al., 2009).
- 12–25-month-olds recognised a social partner and learned novel patterns via Face Time, but not pre-recorded videos (Myers et al., 2017).



Interested

If you are interested in developmental psychology and wish to learn more about myths that are debunked with more recent research:

Hupp, S., & Jewell, J. (2015). Great myths of child development (1st ed.). John Wiley & Sons.



What is this module about?

- This module introduces you to rigorous and transparent scientific practices.
- Using modern techniques, you will learn how to collect, present and analyse data.



How does it work?



Pre-recorded lectures



Readings



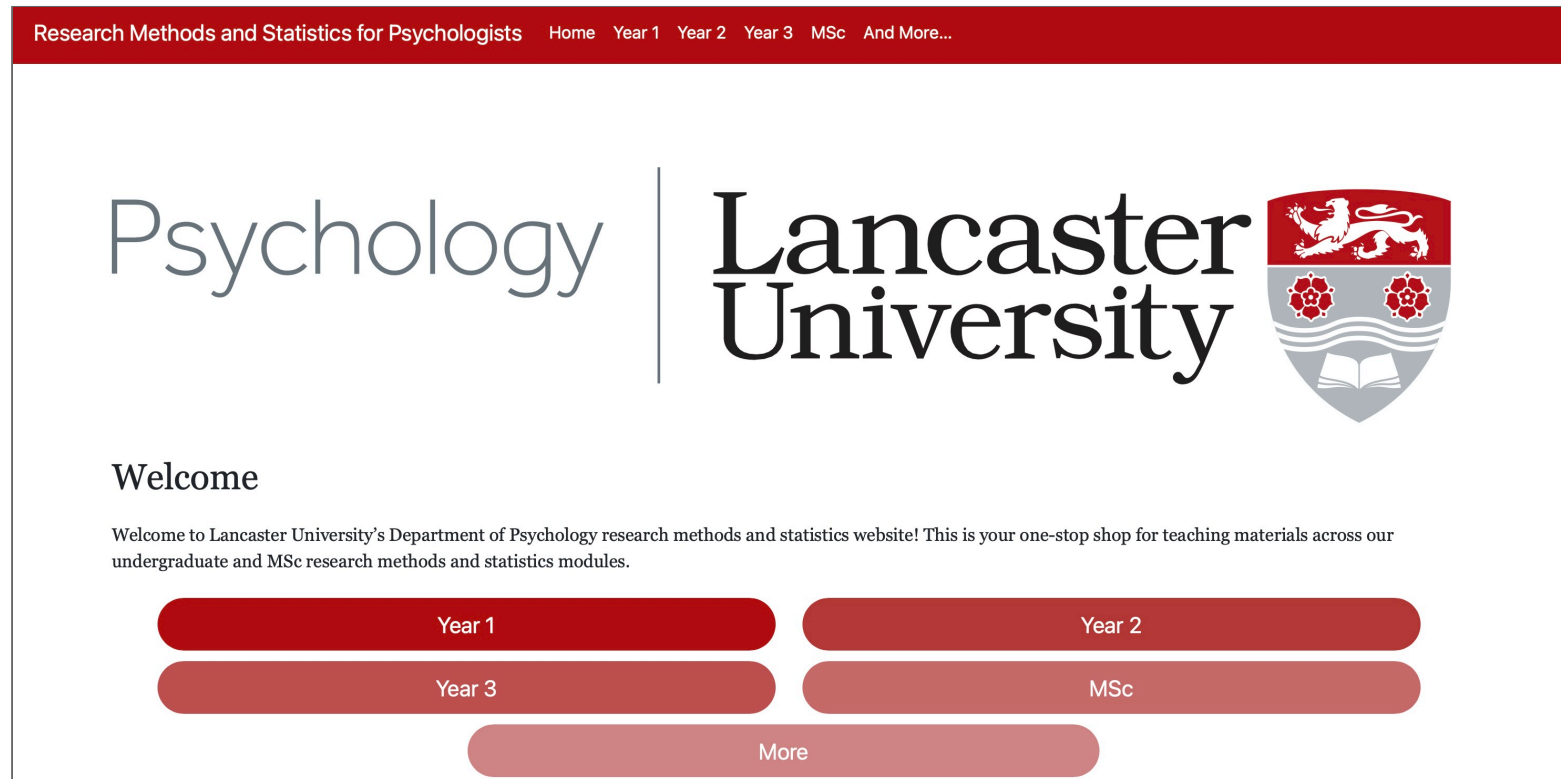
Practice tasks



Labs

What do you need?

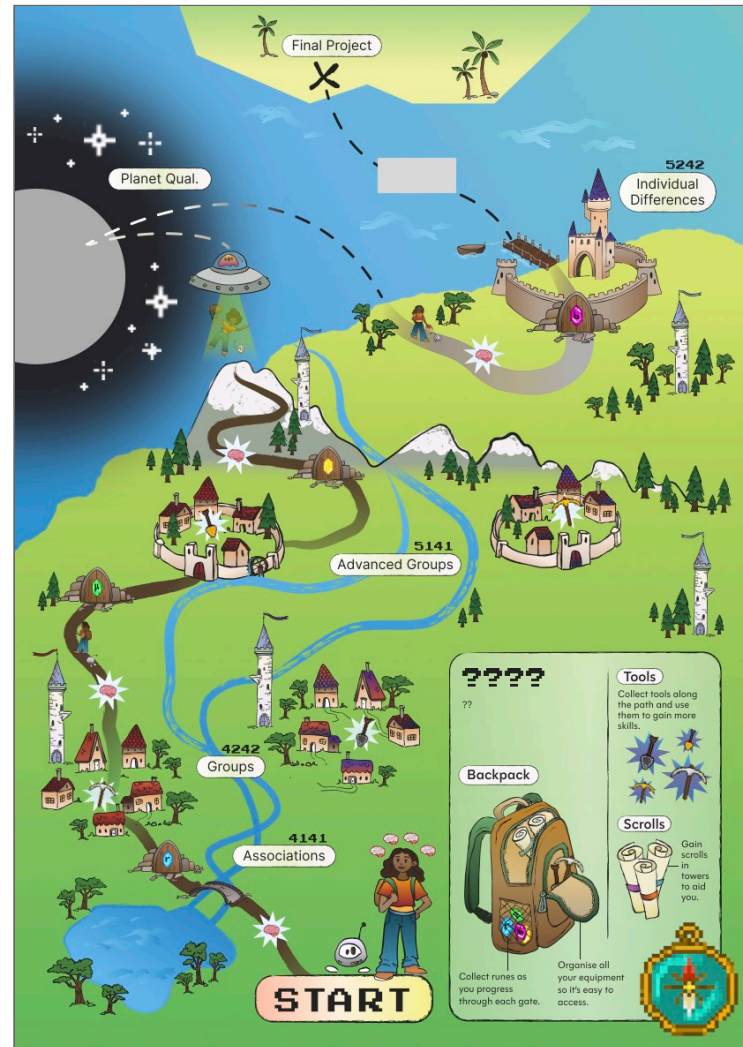
- The research methods and stats handbook <https://lu-psy-r.github.io>
- The R Studio server: <https://ps-rstudio.lancaster.ac.uk/> You need to be on campus or using the **VPN** to login to the R Studio server.
- Library resources -> **Resource list**



Which topics will we cover?

- Describe the research process, from concept design to post-publication
- Use specialist software to build relevant tools (e.g., questionnaires) for data collection.
- Report data in text, tables and visualisations, following relevant conventions and good practice.
- Identify statistical tests that are appropriate for a given design.
- Perform statistical analyses using a statistical software package, interpret the output and report the results following relevant conventions and good practice.
- Explain open science practices.
- Apply evidence-based reasoning and critically engage with empirical research.
- Participate in research both as an experimenter and a research study participant.

How does this module relate to the rest of the curriculum?



How will you be assessed?

Level/Year	Term	Module	Exam	Research report
L4/Y1	1	PSYC4141	50%	35% - 1500 words
L4/Y1	2	PSYC4242	50%	35% - 1500 words
L5/Y2	1	PSYC5141	60%	40% - 2750 words
L5/Y2	2	PSYC5242	60%	40% - 2500 words (excl. intro)
L6/Y3	1 and 2	PSYC6041	dissertation	

How to ask for help

- **Ask your friends on your table** — working together during labs is encouraged
- **Ask a GTA or lecturer** - there are no stupid questions!
- **Ask on the Q & A Forum on Moodle** - if you are wondering other are probably too
- **Ask during the live online Q & A session** - link is on Moodle